Capacity-Dependent Effects of Data Selection for Reasoning

When should LLMs learn from easy versus hard supervision?

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arxiv.org/abs/2608.13721

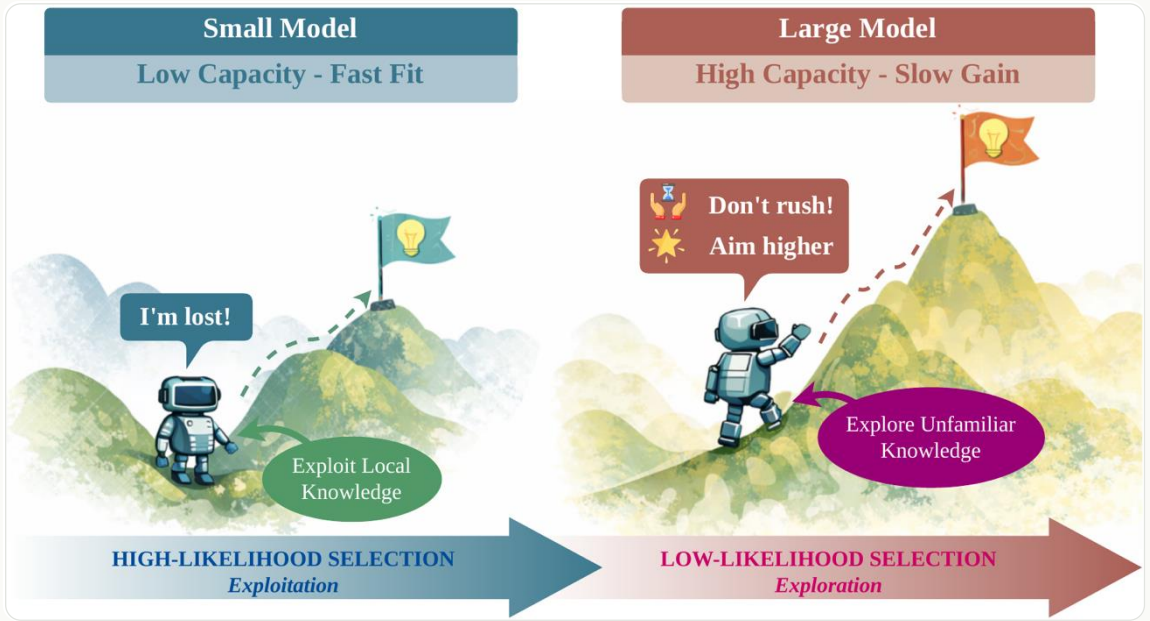


Figure 1: Capacity-dependent learning under likelihood-based data selection.

Math-specific pretraining laid the foundation for reasoning

Jun 2022 · Google

Minerva

38.5B tokens

Continued pretraining of PaLM on arXiv papers and math web pages; no tools.

540B: MATH **50.3%** (maj@64) · 33.6% single sample

Oct 2023 · EleutherAI

Llemma

55B tokens

Code Llama continued on Proof-Pile-2: arXiv 29B · web incl. OpenWebMath 15B · math code 11B.

Open data and weights (7B, 34B); Llemma 7B beats Minerva 8B on MATH

Feb 2024 · DeepSeek

DeepSeekMath

120B

120B math tokens mined from Common Crawl, mixed with natural language and code; 7B model.

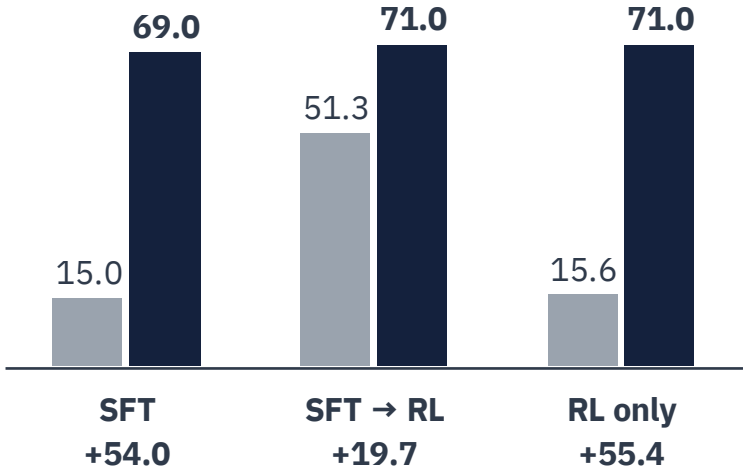
7B: MATH **51.7%** (no tools, no voting) · 60.9% self-consistency@64

Trend: larger, more targeted, increasingly open math corpora. A 7B model trained on curated math data matches a 540B model from two years earlier.

Sources: Lewkowycz et al. (2022). Solving quantitative reasoning problems with language models. NeurIPS. · **Azerbayev et al. (2024)**. Llemma: An open language model for mathematics. ICLR. · **Shao et al. (2024)**. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. arXiv:2402.03300.

Post-training became a powerful, data-efficient route to math reasoning

MetaMath Sep 2023 · SFT	395K bootstrapped GSM8K/MATH questions → **7B: GSM8K 66.5%**, +11.5 over same-size SOTA
OpenAI o1 Sep 2024 · RL	Large-scale RL over long chains of thought → **AIME 2024: 74%** single sample vs GPT-4o 12%
DeepSeek-R1 Jan 2025 · RL + distill	R1-Zero: pure RL on DeepSeek-V3-Base, no SFT; reasoning then **distilled into small dense students**
OpenThoughts Jun 2025 · open SFT, small models	1.2M QwQ-32B traces → **OpenThinker3-7B: AIME25 53.3%**, +15.3 over R1-Distill-Qwen-7B. A stronger model is not always a better teacher.



SFT: Qwen2.5-7B-Instruct → OpenThinker3-7B
SFT → RL: R1-Distill-Qwen-7B → AceReason-7B
RL only: DeepSeek-R1-Zero, start → end of RL

Sources: Yu et al. (2024). MetaMath: Bootstrap your own mathematical questions for large language models. ICLR. · **OpenAI (2024)**. Learning to reason with LLMs. · **Guo et al. (2025)**. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. arXiv:2501.12948. · **Guha et al. (2026)**. OpenThoughts: Data recipes for reasoning models. ICLR. · **Liu et al. (2026)**. AceReason-Nemotron 1.1: Advancing math and code reasoning through SFT and RL synergy. ICLR.

Strong teachers generate reasoning Smaller students learn to imitate

1	2	3	4	5	6
Sample math problems x	Teachers generate candidate traces	**Filter / select** one response	Build SFT dataset $\mathcal{D}$	Fine-tune the student	Evaluate on held-out reasoning

Supervised fine-tuning objective

$$\min_{\theta} \; - \sum_{(x,y) \in \mathcal{D}} \log \, \pi_\theta(y \mid x)$$

$\boldsymbol{x}$ — math question · $\boldsymbol{y}$ — teacher reasoning response

$\boldsymbol{\pi}$ — student policy · $\mathcal{D}$ — the selected training data

The objective is fixed. Step 3 alone decides what the student should imitate.

The final answer is not the only criterion — the distribution matters too

Question:

Solve $x^2 - 5x + 6 = 0$

Answer A · concise, familiar

$(x - 2)(x - 3) = 0$, so $x = 2$ or 3

Answer B · longer, less familiar

Move the constant: $x^2 - 5x = -6$. Complete the square by adding $(5/2)^2 = 25/4$ to both sides: $(x - 5/2)^2 = 1/4$. Take square roots: $x - 5/2 = \pm 1/2$, so $x = 3$ or $x = 2$.

Both correct. Which is better?

2022 · generation

SLiC

Sequence **likelihood affects task performance** in text generation.

Zhao et al.

2024 · preferences

On-policy data

Training on data **likely under the model** improves fine-tuning results.

Tajwar et al.

2025 · SFT

GRAPE

Pick the response the **target model finds most likely**: up to +13.8% avg over distilling from the strongest model.

Zhang, Dai & Peng

2025 · math CoT

Shape of Thought

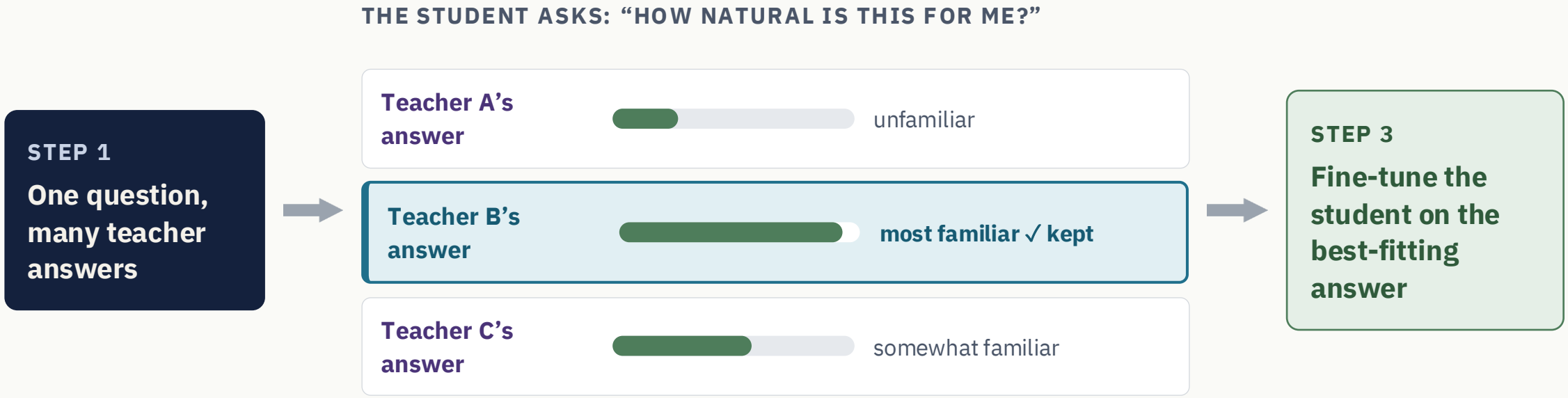
Model CoTs, **even with wrong answers**, beat human-written ones; paraphrasing human traces toward the model helps.

Chandra et al.

It depends on the student: good supervision is correct **and** learnable. **Our question:** is closer always better?

Sources: Zhao et al. (2022). Calibrating sequence likelihood improves conditional language generation. arXiv:2210.00045. · **Tajwar et al. (2024)**. Preference fine-tuning of LLMs should leverage suboptimal, on-policy data. ICML. · **Zhang, Dai & Peng (2025)**. The best instruction-tuning data are those that fit. NeurIPS. · **Chandra et al. (2025)**. Shape of Thought: When distribution matters more than correctness in reasoning tasks. arXiv:2512.22255.

GRAPE selects the responses that best fit the student



Why: teachers differ in style and complexity; answers close to the student should be easier to absorb.

Cost: one scoring pass by the student per candidate; no extra training signal needed.

Source: Zhang, Dai & Peng (2025). The best instruction-tuning data are those that fit. NeurIPS.

Prior research offers two opposing selection principles

PERSPECTIVE A

High-likelihood data is better

Motivated by GRAPE, on-policy distillation, student-distribution alignment, learnability and stable optimization.

Hypothesis: supervision closer to the student policy produces more effective learning.

Learnability

vs

Novelty

PERSPECTIVE B

Hard examples are better

Motivated by hard-example mining, active learning, curriculum research, informative or surprising examples.

Hypothesis: examples far from the student policy can teach capabilities it does not yet have.

Caveat. Low likelihood ≠ useful mathematical difficulty. It can also reflect response style, length or other mismatch.

Sources: Zhang, Dai & Peng (2025). The best instruction-tuning data are those that fit. NeurIPS. · **Agarwal et al. (2024)**. On-policy distillation of language models: Learning from self-generated mistakes. ICLR. · **Lewis (1995)**. A sequential algorithm for training text classifiers. SIGIR Forum. · **Shrivastava, Gupta & Girshick (2016)**. Training region-based object detectors with online hard example mining. CVPR. · **Robinson et al. (2021)**. Contrastive learning with hard negative samples. ICLR. · **Paul, Ganguli & Dziugaite (2021)**. Deep learning on a data diet: Finding important examples early in training. NeurIPS. · **Lin et al. (2024)**. Not all tokens are what you need for pretraining. NeurIPS.

The best training data may depend on the learning context

Model
Capacity · initialization · existing capabilities

Data
Size · difficulty · diversity · teacher–student mismatch

Data selection

Optimization
Learning stage · epochs · learning rate · compute

Objective
Downstream task · evaluation distribution

PRIOR AXES

Learning stage: curriculum & self-paced learning (Bengio 2009; Kumar 2010; Weinshall 2018).
Data size: keep easy examples when data is scarce, hard when abundant (Sorscher 2022).

THIS WORK

Adds **student capacity** × **training duration** for student-likelihood response selection.

Sources: Bengio et al. (2009). Curriculum learning. ICML. · **Kumar, Packer & Koller (2010)**. Self-paced learning for latent variable models. NeurIPS. · **Weinshall, Cohen & Amir (2018)**. Curriculum learning by transfer learning: Theory and experiments with deep networks. ICML. · **Sorscher et al. (2022)**. Beyond neural scaling laws: Beating power law scaling via data pruning. NeurIPS.

2 · RESEARCH MOTIVATION

Is there a universal winner between high and low likelihood?

RQ1

Does high-likelihood supervision always win?

Compare five student sizes over short and long training.

RQ2

Why do students respond differently?

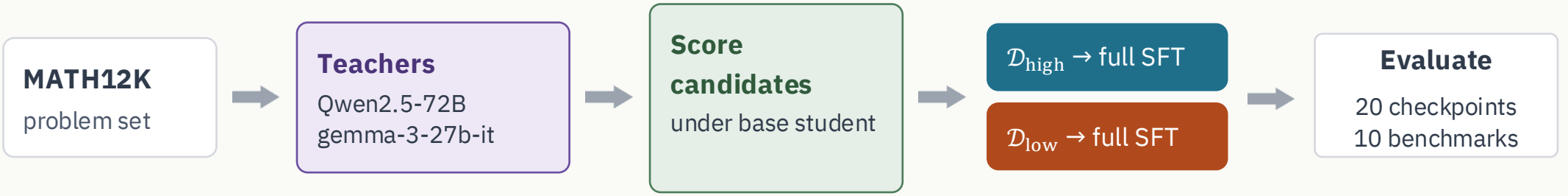
Track generated outputs under the teacher and the original student.

RQ3

What mechanism governs transfer?

Data span × knowledge gap × feasible student capacity.

Controlled response selection isolates the effect of student likelihood

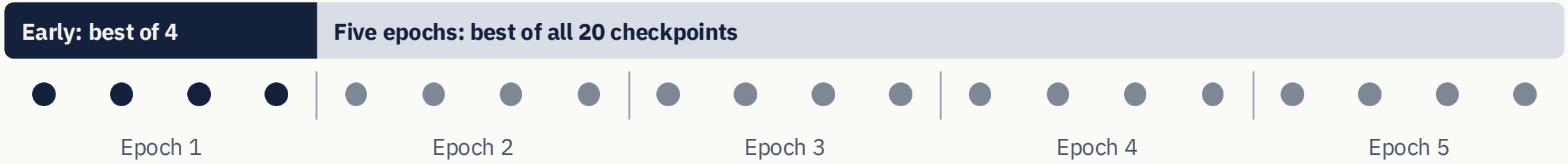


STUDENTS

Size	Model
1.5B	Qwen/Qwen2.5-1.5B
3B	Qwen/Qwen2.5-3B-Instruct
4B	Qwen/Qwen3-4B
7B	Qwen/Qwen2.5-Math-7B
8B	Qwen/Qwen3-8B

Sources: Hendrycks et al. (2021). Measuring mathematical problem solving with the MATH dataset. NeurIPS. · **Qwen (2025)**. Qwen2.5 technical report. · **Yang et al. (2024a)**. Qwen2.5-Math technical report: Toward mathematical expert model via self-improvement. · **Yang et al. (2025)**. Qwen3 technical report. · **Gemma Team (2025)**. Gemma 3 technical report.

Evaluating both early learning and long-run performance



PROTOCOL

- Pass@1, temperature 0.6
- Scores are **best-checkpoint envelopes** per benchmark, not final checkpoints

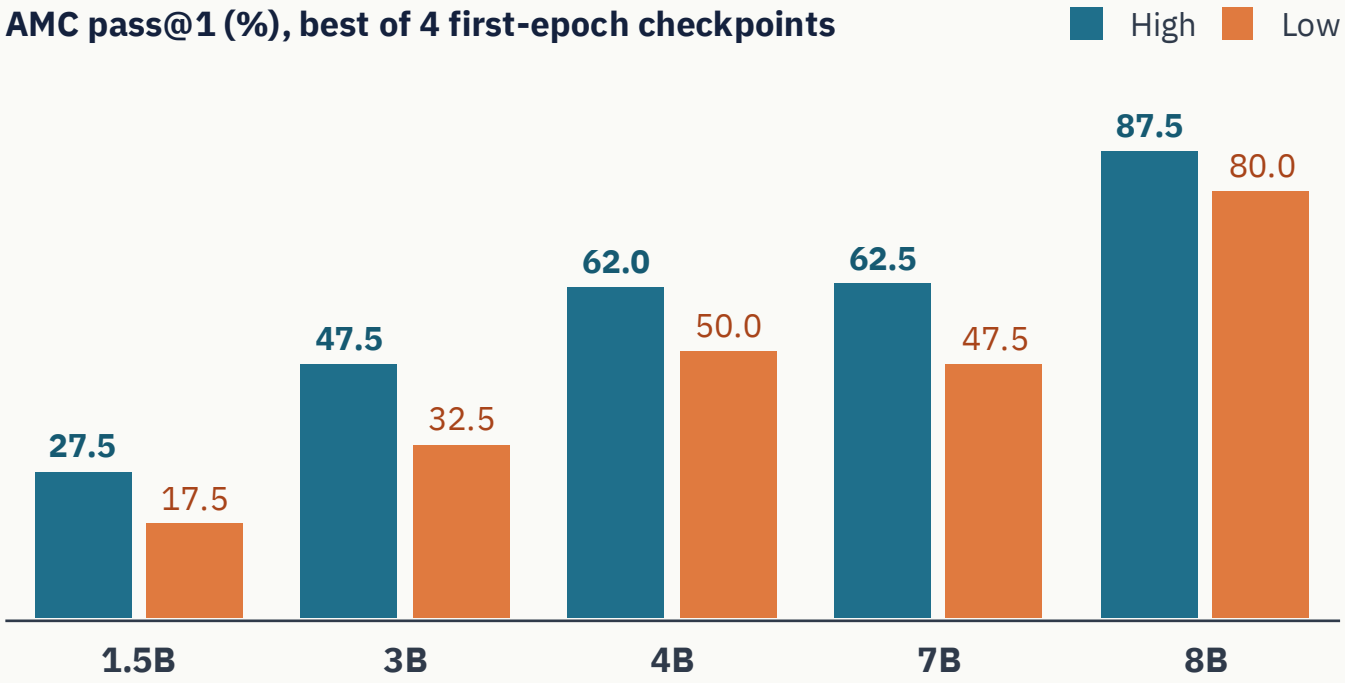
TEN BENCHMARKS

AIME24 · AMC · CHMATH · Gaokao · GPQA · GradeSchool · KAOYAN · MATH500 · Minerva · OlympiadBench

Sources: AIMO (2024). AIMO validation sets: AIME 2024 and AMC. · **Wei et al. (2023)**. CMATH: Can your language model pass Chinese elementary school math test? · **Zhang et al. (2023)**. Evaluating the performance of large language models on GAOKAO benchmark. · **Rein et al. (2024)**. GPQA: A graduate-level Google-proof Q&A benchmark. COLM. · **Hendrycks et al. (2021)**. Measuring mathematical problem solving with the MATH dataset. NeurIPS. · **Lewkowycz et al. (2022)**. Solving quantitative reasoning problems with language models. NeurIPS. · **He et al. (2024)**. OlympiadBench: A challenging benchmark for promoting AGI with olympiad-level bilingual multimodal scientific problems. ACL. · **Ye et al. (2025)**. LIMO: Less is more for reasoning. COLM.

High-likelihood data consistently wins during early training

AMC pass@1 (%), best of 4 first-epoch checkpoints

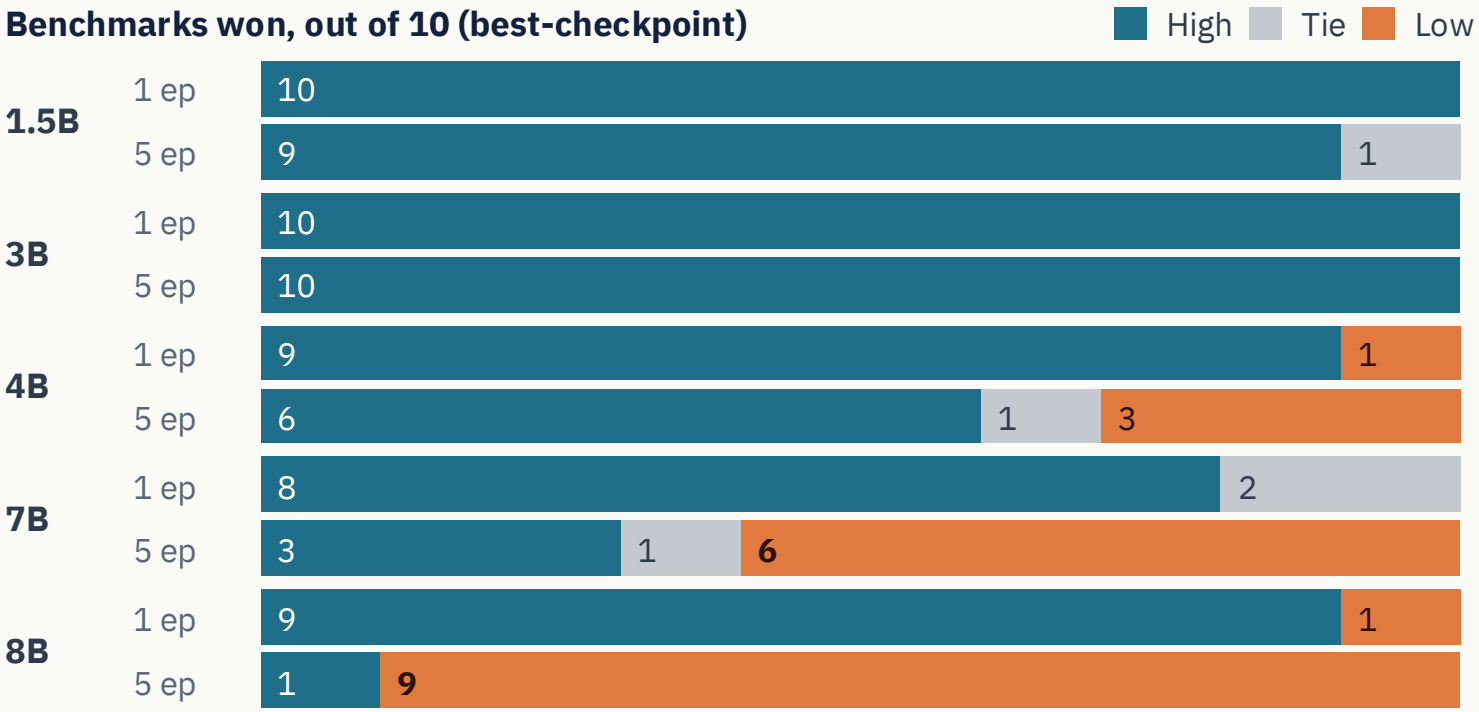


ALL 10 BENCHMARKS, EPOCH 1

Student	High wins
1.5B	10 / 10
3B	10 / 10
4B	9 / 10
7B	8 / 10 (2 ties)
8B	9 / 10

Observation 1. High-likelihood data yields faster early gains, especially for smaller students.

After five epochs, low likelihood wins for the larger students



EXAMPLES, TABLE 2

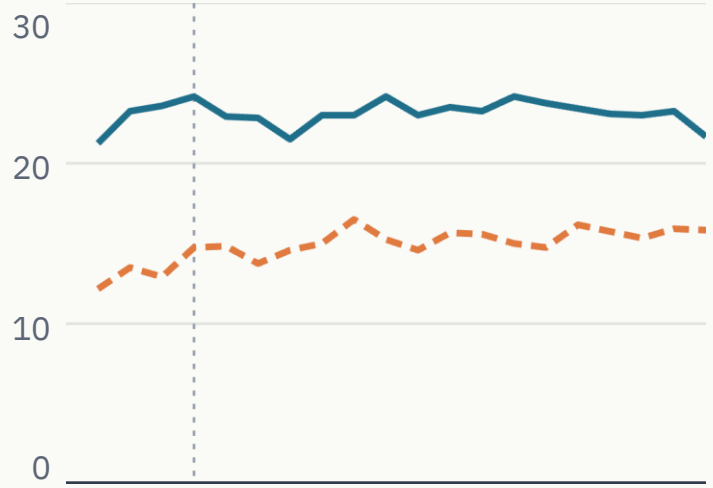
Case	High	Low
1.5B AMC	35.0	22.5
3B AMC	55.0	47.5
7B GPQA	39.39	44.95
7B MATH	79.00	80.06
8B AMC	90.0	92.5
8B MATH	92.0	93.0

Not uniform: 8B still prefers High on Gaokao (74.68 vs 67.09).

Observation 2. For high-capacity models, high likelihood peaks early, while low likelihood improves later and can surpass it.

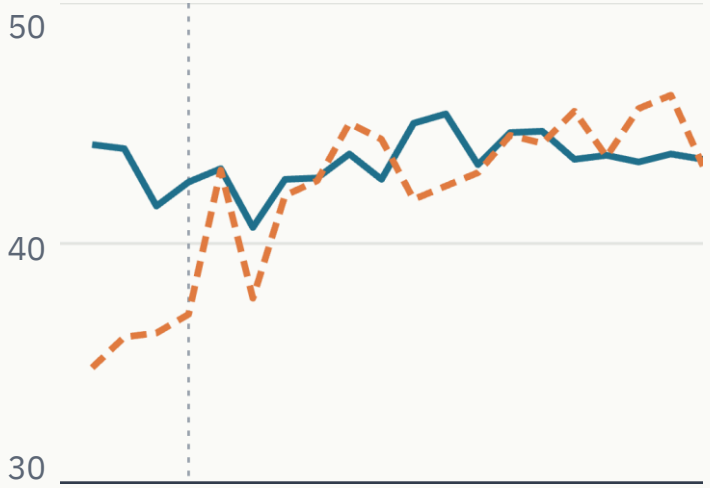
High likelihood fits quickly; low likelihood can gain more eventually

1.5B · Fast-Fit, no catch-up



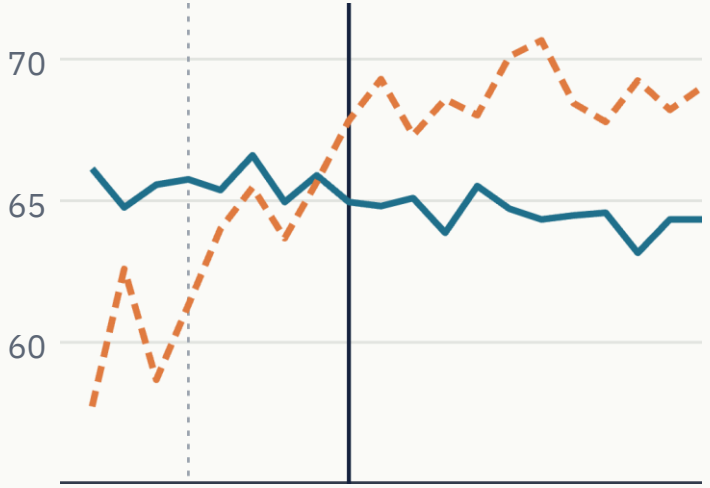
High 24.16 by epoch 1; Low's best is only 16.50.

7B · Unstable Slow-Gain



Low peaks at 46.19 > High's 45.41, but leads at only 5 of 20 checkpoints.

8B · Sustained Slow-Gain



Early → best: High +0.48, Low +8.07 (to 70.69).

High (solid) Low (dashed) · Simple average of 10 benchmarks, pass@1, every checkpoint

Small models can fail to absorb unfamiliar reasoning

Question: There is a sequence $a_1, a_2, \ldots, a_{2023}$ such that $\tan(2023x) = \cdots$. What is a_{2023}?

Qwen2.5-1.5B	Qwen2.5-3B-Instruct	Qwen2.5-Math-7B
There is a sequence $a_1, a_2, \ldots, a_{2023}$ such that $\tan(2023x) = \cdots$. What is a_{2023}?	Let $P(x) = \tan 2023x$. Then	Let $t = \tan x$. Then we have...
	$\tan(2023x) = \dfrac{\tan(2022x) + \tan x}{1 - \tan(2022x)\tan x}$	We know that
		$\tan(A + B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$
$\boxed{2023}$ $\boxed{2023}$ $\boxed{2023}$ $\boxed{2023}$ $\cdots$	$\tan(2023x) = \dfrac{\tan(2022x) + \tan x}{1 - \tan(2022x)\tan x}$	$\cdots$
	$\cdots$	Final answer: $\boxed{1}$.

1.5B: repeats the question, then loops on an incorrect answer — shallow imitation.

3B: begins valid steps, then gets lost repeating them.

7B: completes the reasoning and reaches the correct answer.

Observation 3. Small model: repetition, shallow copying, incomplete reasoning. Larger model: delayed but meaningful reasoning improvement.

Selection effectiveness depends on capacity × training time

	Short training	Long training	Regime
Small 1.5B · 3B · 4B	**High** strongly favored	**High** remains favored	Fast-Fit, no catch-up
Intermediate 7B	**High** favored	Low reaches higher peaks; lead is checkpoint-sensitive	Fast-Fit, unstable Slow-Gain
Large Qwen3-8B · Llama-3.1-8B	**High** often favored	**Low** can outperform, persistently	Fast-Fit, sustained Slow-Gain

Patterns in the evaluated models — not a universal scaling threshold. Model families and specializations also differ across sizes.

Takeaway: Choose supervision difficulty to match the student's capacity and compute budget.

Learning dynamics reveal what the student actually learns

Accuracy alone can't tell **meaningful movement toward the teacher** from **mere drift away from the starting point**.

Fine-tuned student π_t

generates answers y at each checkpoint t

Teacher view: is the student moving *toward* the teacher?

$$\mathcal{L}_T(t) = -\,\mathbb{E}_x\,\mathbb{E}_{y\sim\pi_{\theta_t}(\cdot\,|\,x)}\left[\log \pi_T(y\,|\,x)\right]$$

Base-student view (π_0): how far has it moved *away* from where it started?

$$\mathcal{L}_0(t) = -\,\mathbb{E}_x\,\mathbb{E}_{y\sim\pi_{\theta_t}(\cdot\,|\,x)}\left[\log \pi_{\theta_0}(y\,|\,x)\right]$$

Low-likelihood data pulls capable students toward the teacher

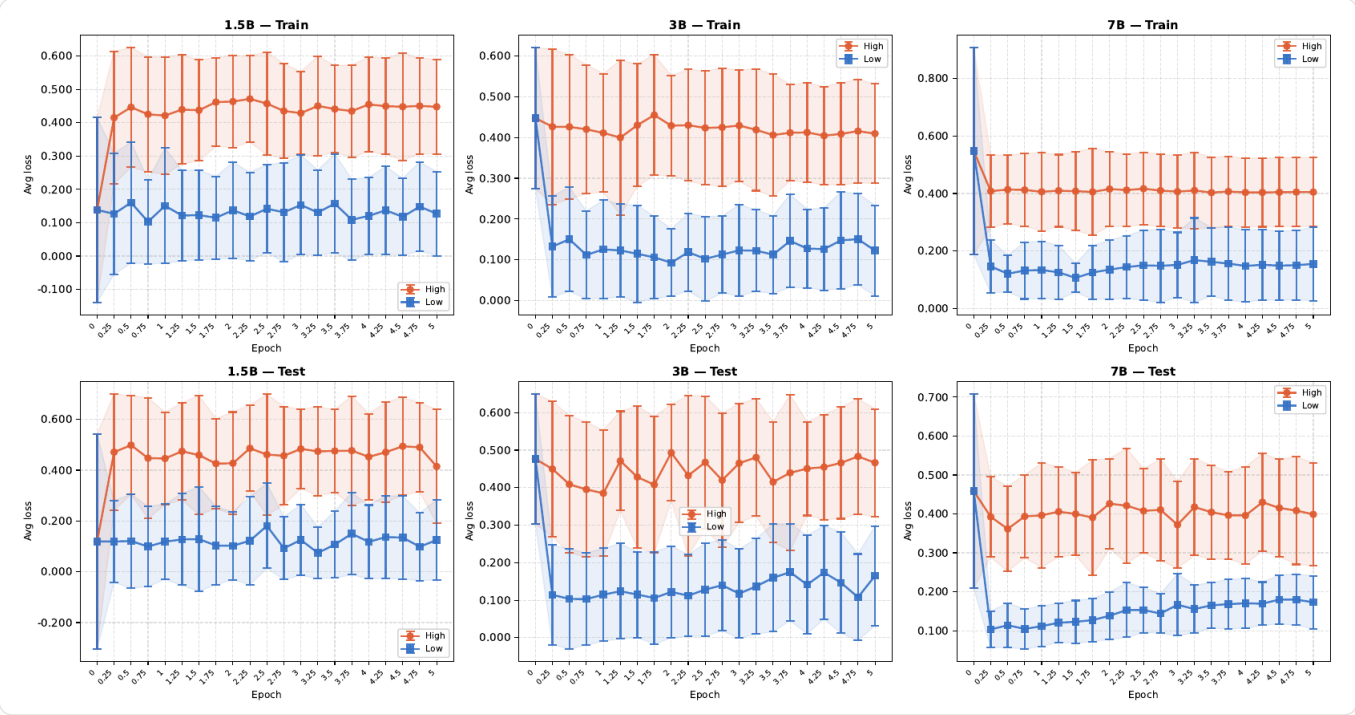


Figure 2: Loss under teacher model. Lower = closer to teacher.

1.5B: High training raises teacher loss (drifts away); Low leaves it roughly unchanged.

3B: High stays flat; Low **decreases** teacher loss.

7B: both decrease; Low reaches the lower loss.

Observation 4. Low-likelihood data moves larger models closer to the teacher but fails to do so for small models.

Low-likelihood training pushes students beyond their initial distribution

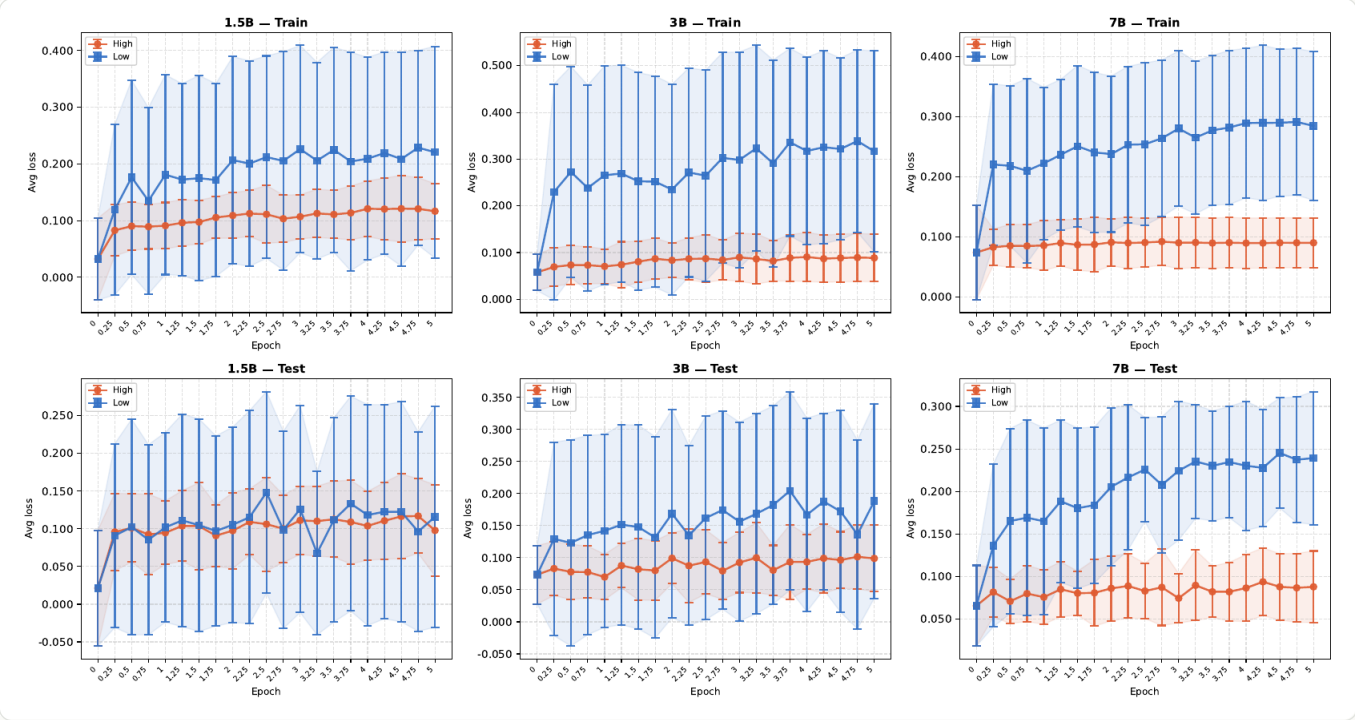


Figure 3: Loss under original student model. Higher = farther from the base student.

> **1.5B:** Low gives a broader, more divergent pattern that still overlaps the initial range.

> **3B:** clearer shift; High–Low separation grows.

> **7B:** smaller variance than 3B — a more stable, focused shift.

Shift alone is not proof of better reasoning.

Observation 5. As size grows, behavior is more stable, generalization improves, and the High/Low distinction becomes more pronounced.

Teacher knowledge is filtered by data coverage and student capacity

Knowledge gap
$\Delta \mathbf{w} = \mathbf{w}_* - \mathbf{w}_0$

→

Exposed by data
$\mathrm{span}\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$

→

Representable
feasible subspace $\mathcal{F}$

→

Learned
$\widehat{\mathbf{w}} - \mathbf{w}_0$

SETTING

Linear teacher and student classifiers:

$$h^*(\mathbf{x}) = \mathbf{1}\{\mathbf{w}_*^\top \mathbf{x} \geq 0\}, \qquad h(\mathbf{x}) = \mathbf{1}\{\mathbf{w}^\top \mathbf{x} \geq 0\}$$

Soft teacher labels $y_i = \sigma(\mathbf{w}_*^\top \mathbf{x}_i)$

Gradient flow on the cross-entropy loss:

$$L(\mathbf{w}) = -\tfrac{1}{n}\sum_{i=1}^{n}[y_i \log \sigma(\mathbf{w}^\top \mathbf{x}_i) + (1-y_i)\log(1-\sigma(\mathbf{w}^\top \mathbf{x}_i))]$$

Scope

A simplified binary-classification model of distillation — not a nonlinear LLM. Low capacity = student confined to a subspace $\mathcal{F} \subseteq \mathbb{R}^d$.

Training data reveal only part of the teacher's knowledge

High-capacity student. Let $P_{\mathbf{X}}$ project onto $\mathrm{span}\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$. As $\tau \to \infty$, $\mathbf{w}(\tau) \to \hat{\mathbf{w}}$ with

$$\hat{\mathbf{w}} = \mathbf{w}_0 + P_{\mathbf{X}}(\mathbf{w}_* - \mathbf{w}_0)$$

- The learned change is the **projection of the knowledge gap** onto the data span
- Directions orthogonal to the data are **never updated**; the student stays anchored to $\mathbf{w}_0$ there

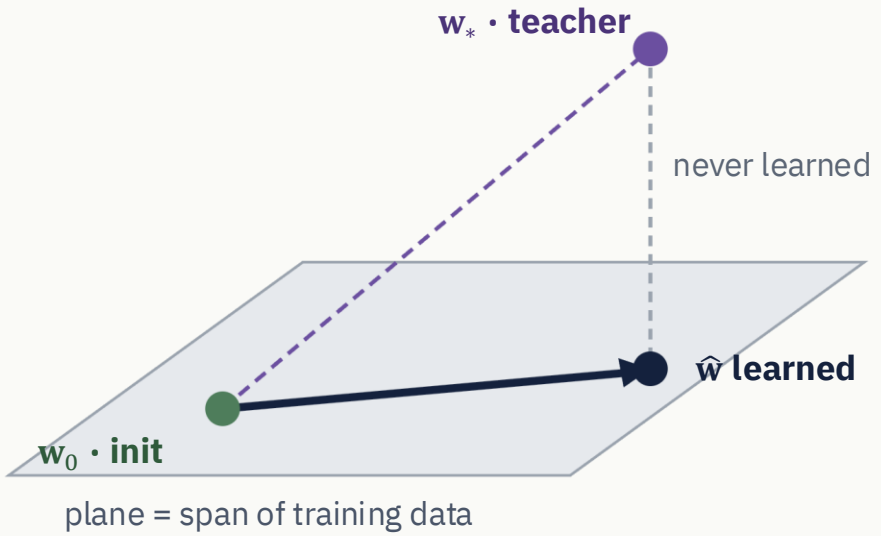


Figure 4: Distillation as projection. The student moves from $\mathbf{w}_0$ only within the data span, reaching $\hat{\mathbf{w}}$; the teacher's off-plane component is never learned.

Low-capacity students learn only representable teacher knowledge

Let $\mathbf{X}_{\mathcal{F}}$ be the data span projected onto $\mathcal{F}$, and $P_{\mathbf{X}_{\mathcal{F}}}$ its projector.
As $\tau \to \infty$:

$$\hat{\mathbf{w}} = \mathbf{w}_0 + P_{\mathbf{X}_{\mathcal{F}}}(\mathbf{w}_* - \mathbf{w}_0)$$

Takeaway 2. Low-capacity distillation needs data that expose teacher knowledge *and* place it within the student's representable subspace.

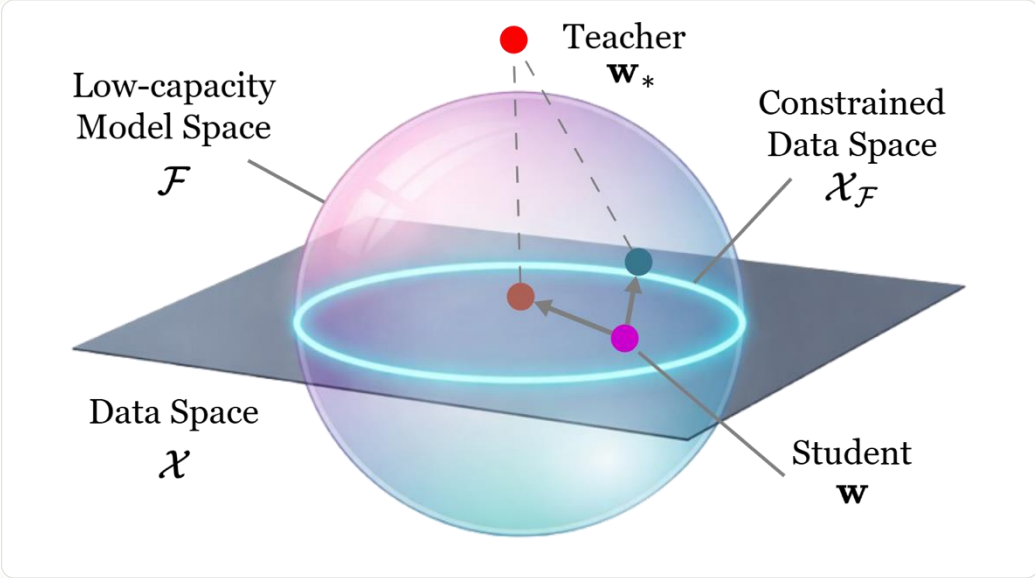


Figure 5: High-capacity students move toward the **brown** point (optimal solution); low-capacity students only toward the **teal** point (representable part – suboptimal solution). The remainder is unrepresentable.

6 · THEORETICAL EXPLANATION

Why low-likelihood supervision can fail for small models

Low-capacity student

1. Low-likelihood data expose unfamiliar teacher behavior
2. Much of the gap lies outside the feasible space $\mathcal{F}$
3. The supervision cannot be fully assimilated
4. Training yields shallow copying or repetition
5. More difficulty ≠ better reasoning

Higher-capacity student

1. Low-likelihood data expose unfamiliar teacher behavior
2. More of the gap is representable
3. Continued optimization lets it adapt
4. It approaches the teacher distribution
5. Accuracy can surpass the high-likelihood baseline

The trajectories **motivate** this reading; the theorems establish a mechanism **in the simplified linear model** — not an exact account of transformer training, nor a proof of Fast-Fit / Slow-Gain dynamics.

CONCLUSION

There is no universally optimal likelihood-based selection rule

1

Familiarity matters

High-likelihood supervision gives faster, more stable early gains — across every student size.

2

Capacity matters

Low-likelihood data pays off only when the student can represent and absorb unfamiliar teacher knowledge.

3

Duration matters

Short budgets favor easy-to-learn data; larger models trained longer can gain from challenging data.

The best reasoning data is not always the easiest or the hardest — it is the data the student can *productively learn from*.

Effective data selection is a function of student capacity, learning dynamics and training budget — not sample difficulty alone.

Thank you for your attention!

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